

Model review and revisions

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Tommaso De Santo

I reviewed the full model and calibration: preferences, borrowing rules, demographic accounting, equilibrium conditions, empirical measurements, targets, and the estimation procedure.

- Replaced the minimum-housing requirement with a first-child increase in the housing expenditure share.
- Allowed diminishing benefits from children at home, distinguishing them from children ever born.
- Separated children leaving home from adult household entry, and tied entrant assets to estates.
- Rebuilt the first-birth housing evidence and recalibrated the revised model.

Household decisions

Overlapping generations of households differ in age, earnings, wealth, housing tenure, and family size.

- During fertile ages, households choose whether to attempt a birth; success becomes less likely with age.
- They choose consumption, saving, housing size, and whether to rent or own; renters pay rent, while owners make a down payment and face selling costs.
- Persistent earnings risk and borrowing limits shape saving; a first-birth fixed cost and taste shocks shape fertility.
- Households retire, face mortality risk, and value bequests.

The interaction of housing costs and available wealth affects both birth timing and tenure choice.

Children and housing demand

Households consume nonhousing goods c and housing services s , and derive utility from the m children living at home:

$$u(c, s, m) = \frac{\left[\frac{A(m)c^{\alpha(m)}s^{1-\alpha(m)}}{e(m)} \right]^{1-\sigma}}{1-\sigma} + \psi m^{1-\gamma}.$$

The equivalence scale $e(m)$ captures the consumption cost of a larger family; σ governs risk aversion. The benefit from children is zero at $m = 0$, with ψ setting its level and γ its curvature.

Having children also raises the share of expenditure devoted to housing:

$$\alpha(m) = \alpha_0 - \Delta\alpha \mathbf{1}\{m > 0\}.$$

α_0 is the childless consumption share. The first child raises the housing share by $\Delta\alpha$; additional children leave that share unchanged.

For given spending and family size, $A(m)$ offsets the utility effect of the share change at a fixed reference rent.

Demographic and market equilibrium

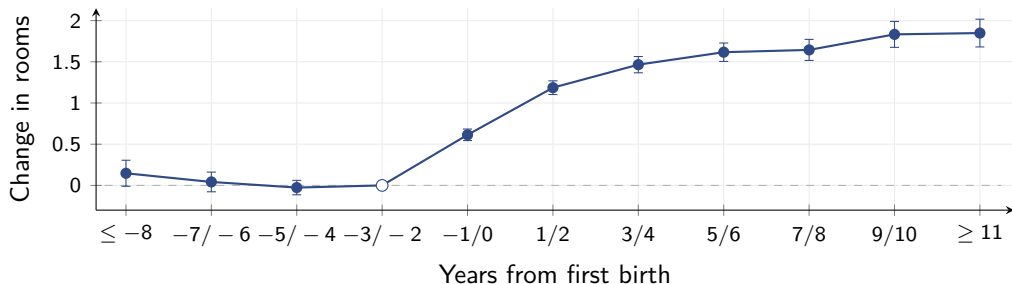
Children leaving home and new adult households entering the economy are separate transitions.

- Children leave dependency independently; adult entry follows earlier births with a lag.
- Positive estates finance fixed positive entrant assets; any remainder leaves the economy.
- The house price clears aggregate housing demand against housing supply.
- Equal retiree pensions are financed by a balanced payroll tax.

A stationary equilibrium has constant prices, allocations, and cohort structure, with demographic reproduction and market clearing.

Housing around the first birth

I rebuilt the PSID event study using interview dates and household status before the first birth.



At years 3–4, rooms rise by **1.465** (SE 0.050), ownership by **26.886 pp**, and recent moving falls by **21.147 pp**.

Reference person or spouse before birth. Baseline: years $-3/-2$; person and year effects; survey weights; 95% pointwise intervals.

Calibration strategy and inputs

I recalibrated a stationary economy representing 2007, using national measures of housing, earnings, wealth, and fertility.

Input	Value	Measurement
Mean occupied rooms	5.729	AHS 2007
Pension / working-household earnings	0.229	CPS ASEC, income 2007
Annual child-directed estates / wealth	0.729%	SCF 2007 mortality proxy
Annual depreciation / property tax	1.416% / 1.060%	National housing inputs

- Fixed inputs include earnings and demographic processes, housing depreciation and taxes, the interest rate, and mortgage financing.
- At each candidate parameter vector, I solve household choices and housing-market equilibrium, adjusting ψ separately to match completed fertility of 2.1.
- The objective is a weighted sum of squared gaps for ten moments. Three further moments are retained for validation; the fertility normalization is reported separately.

The weights set working error scales; they are not an estimated joint covariance matrix.
The first-birth housing comparison uses a model proxy for the empirical event study.

2007 stationary calibration

Moment	Target	Model
Completed fertility (normalization)	2.100	2.100
Childlessness, ages 40–44 (%)	19.828	20.106
Exactly one child among mothers, ages 40–44 (%)	21.366	20.942
Mean first-birth age (years)	25.976	25.933
First births at age 30+ (%) [†]	24.928	22.365
Wealth / earnings	6.927	6.326
Annual child-directed bequest flow / wealth (%)	0.729	0.706
Older wealth/income p90/p50 [†]	3.516	3.069
Mean occupied rooms	5.729	5.848
Ownership, ages 30–55 (%)	67.626	65.482
First-birth rooms difference	1.465	1.622
Family rooms difference [†]	0.385	0.353
Recent-parent ownership difference (pp)	12.761	11.955
Children per woman at age 25, capped at three	0.810	0.535

[†]Untargeted validation. Ten scored moments and a separate fertility normalization.

Calibrated parameters

Ten parameters are estimated jointly; ψ is adjusted separately to match completed fertility.

Parameter	Economic interpretation	Value
H_0	Housing supply scale	6.294
β	Annual discount factor	0.963
χ	Owner housing-service premium	1.094
F_1	First-birth fixed utility cost	0.621
κ_1	First-birth taste dispersion	0.176
κ_+	Later-birth taste dispersion	0.332
θ_0	Bequest weight	0.125
$\Delta\alpha$	First-child housing-share loading	0.135
γ	Child-benefit curvature	0.061
κ_T	Tenure-choice dispersion	0.012
ψ	Child-benefit level (normalized)	0.136

Fixed inputs: $\sigma = 2$; $\alpha_0 = 0.733$; annual interest 2%; mortgage-financed share 80%. The fitted values are conditional on these restrictions.

Calibration and identification

I reviewed which empirical variation disciplines each parameter block; the mapping is joint rather than one parameter per moment.

- Housing: mean rooms and first-birth room changes discipline supply scale and the child housing-share loading; ownership levels and parenthood differences discipline the owner premium and tenure tastes.
- Saving: wealth relative to earnings and child-directed bequest flows discipline patience and bequest strength; older wealth dispersion provides an additional check.
- Fertility: childlessness, first-birth timing, the one-child share among mothers, and children per woman at age 25 discipline the first-birth cost, birth tastes, and child-benefit curvature.
- Identification: I checked local moment responses to parameter changes. Some combinations are weakly distinguished, especially child-benefit curvature in a later experimental calibration; parameter confidence intervals are not yet available.

The main remaining fit problem is the number of children among young mothers. Matching the share of mothers alone would hide that miss.

Credit and further calibration

I reviewed the borrowing rules and tested alternative initial-wealth distributions at a common 2% annual real interest rate.

Entrant wealth rule	Renter debt limit (mean annual earnings)
Empirical wealth/earnings ratios drawn independently of earnings	0.25
Zero entrant wealth	0
Nonnegative ratio distribution with the original mean	0

- The three pilot calibrations are complete; an entrant-wealth and credit specification has yet to be selected.
- The next step is to resolve that choice and assess the remaining young-fertility gap.

The stationary solver was reorganized and accelerated to support calibration.